

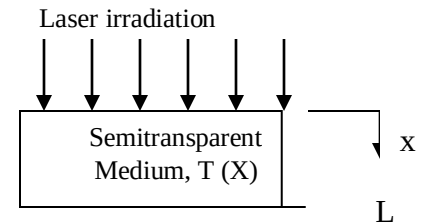
1. It was decided to melt the ice inside a frozen water pipe by passing an electric current I through the pipe wall. The inner and outer wall of the pipe are r_1 and r_2 , and its electrical resistance per unit length is designated as R'_e (Ω/m). The pipe is well insulated on the outside and during melting, the ice (and water) in the pipe remain at a constant temperature T_m associated with the melting process.

(a) Assuming that steady state conditions are reached shortly after application of the current, determine the form of the steady state temperature distribution $T(r)$ in the pipe wall during the melting process.

- (b) Develop an expression for the time t_m required to completely melt the ice. Calculate the time for $I = 100 \text{ A}$, $R'_e = 0.30 \text{ } (\Omega/\text{m})$ and $r_1 = 50 \text{ mm}$.

2. The steady state temperature distribution in a semi-transparent material of thermal conductivity k and thickness L exposed to laser irradiation is of the form,

$T(x) = - (A / ka^2) e^{-ax} + Bx + C$, where A , a , B and C are known constants and k is the thermal conductivity. For this situation, radiation absorbed in the material is manifested by a distributed heat generation term, $Q(x)$.



(a) Obtain expressions for the conduction heat fluxes at the front ($x=0$) and rear ($x=L$) surfaces.

(b) Derive an expression for $Q(x)$.

(c) Derive an expression for the rate at which radiation is absorbed in the entire material, per unit surface area in terms of the known constants for the temperature distribution, thermal conductivity ' k ' and thickness ' L '.

3. Consider a plane wall of thickness L , which is to act as shielding for a nuclear reactor. The inner surface receives gamma radiation that is partially absorbed within the shielding and has the effect of an internally distributed heat source. In particular heat is generated per unit volume within the shielding according to the relation

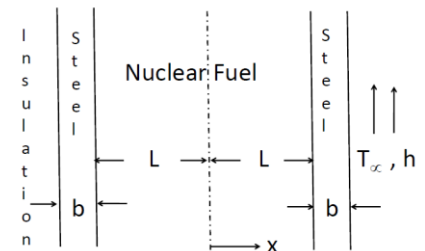
$$q(x) = q_0'' \alpha e^{-\alpha x}$$

Where q_0'' is the incident radiation flux and α is the absorption coefficient of the shielding.

(a) If the inner ($x = 0$) and outer ($x = L$) surfaces of the shielding are maintained at constant temperatures of T_1 and T_2 respectively, find the temperature distribution in the shielding?

(b) Find an expression for the x location in the shield at which the temperature is a maximum.

4. A nuclear fuel element of thickness $2L$ is covered with a steel cladding of thickness b on both sides as shown in the adjoining figure. Heat generated in the nuclear fuel at a rate of q (W/m^3) is removed by a fluid at T_∞ , which adjoins one surface and is characterized by a convection coefficient h . The other surface is well insulated, and the fuel and the steel have thermal conductivities k_f and k_s respectively. Obtain an expression for the temperature distribution $T(x)$ in the nuclear fuel at steady state. Express your answers in terms of q , L , b , k_f , k_s , h and T_∞ . Sketch the temperature distribution for the entire system.



5. The one-dimensional flow of oil between two large parallel plates is to be analyzed at steady state with constant properties and in absence of any significant body forces in the direction of flow. The top plate is moving to the right (in the $+x$ direction), that sustains the flow (pure Couette flow) with the origin of the coordinate system located at the bottom plate. The distance between the two plates is L and there is significant viscous dissipation because of the high viscosity of the oil and the large plate velocity. Note that in this case the dissipation function can simply be expressed as $\mu(dv_x/dy)^2$. The bottom plate is insulated and the top plate is maintained at a constant temperature of T_0 with conduction the prevailing mode of heat transfer.

a) Derive an expression for the velocity distribution in the oil.

b) Derive an expression for the temperature distribution in the oil.

c) Locate the maximum temperature in the flow domain.

d) Evaluate the heat that is to be removed from the upper plate to maintain its temperature constant.